RECITE: Revising Eligibility Criteria Incorporating Textual Evidence

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Abstract

Overly restrictive eligibility criteria contribute to widespread enrollment failure in clinical trials, yet published recommendations for how specific criteria should be broadened remain disconnected from trial design at scale. We introduce RECITE (Revising Eligibility Criteria Incorporating Textual Evidence), a framework for automated discovery, extraction, and implementation of evidence-based eligibility criteria revisions. RECITE leverages protocol amendment documents from ClinicalTrials.gov to construct a dataset of 11,913 curated instances across 5,735 trials, each pairing original criteria with documentary evidence and target revisions. We benchmark eight LLM configurations, with and without retrieval-augmented generation, on a 3,116-instance sample under an LLM-as-judge protocol. The best open-weight model, Gemma 2 9B, achieves 84.7% binary equivalence (normalized score 0.774), while the best closed-weight model, GPT-4o-mini via UCSF Versa, reaches a comparable performance of 85.8%. An agentic literature discovery system identifies 44 paper–trial matches with estimated accrual gains of 25–246%, demonstrating that existing clinical knowledge can be operationalized into actionable trial design recommendations.

CCS Concepts

• **Computing methodologies** → **Natural language processing**; *Machine learning*; • **Applied computing** → *Health informatics*; • **Information systems** → Retrieval models and ranking.

Keywords

Clinical trials, Eligibility criteria, Large language models, Retrieval-augmented generation

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Conference'17, Washington, DC, USA
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ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

Clinical trials are the primary mechanism for evaluating medical interventions, yet enrollment remains a persistent bottleneck. Approximately 80% of trials fail to meet enrollment timelines [8], and insufficient accrual is the single largest cause of trial termination, accounting for 36.7% of all stoppages in a recent analysis of 28,561 discontinued trials [30]. When enrollment shortfalls trigger protocol amendments, the costs compound: 75.8% of protocols undergo at least one amendment, with eligibility criteria modifications representing the most common change type; each Phase III amendment costs a median of $535,000 and adds roughly three months of delay [11, 12]. Overly restrictive eligibility criteria are widely recognized as a contributing factor [25].

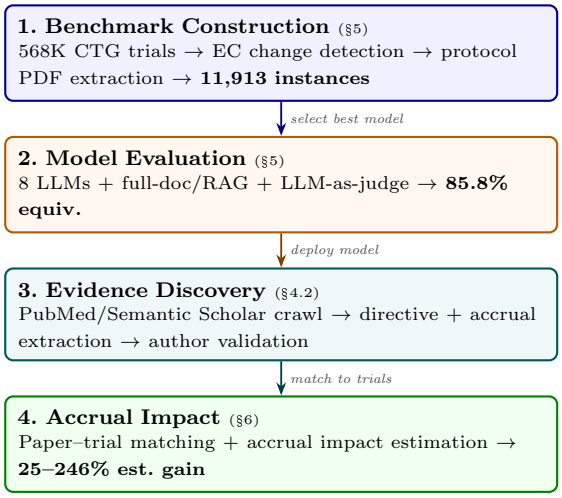

Figure 1: RECITE framework overview. Four stages: benchmark construction from ClinicalTrials.gov protocols (§5), model evaluation via LLM-as-judge, evidence discovery from published literature (§4.2), and estimated accrual impact (§6).

A growing body of work already articulates explicit recommendations for how specific eligibility criteria should be loosened. Governing body directives from the American Society of Clinical Oncology and Friends of Cancer Research (ASCO-Friends) recommend inclusion of patients with controlled brain metastases, prior malignancies, and reduced creatinine clearance thresholds [20]. The U.S. Food and Drug Administration has issued cross-cutting guidance encouraging sponsors to broaden criteria, justify each exclusion, and include

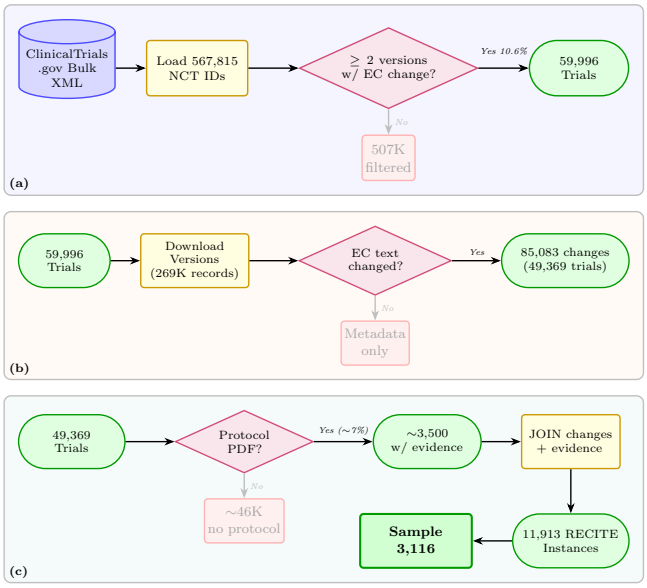

Figure 2: RECITE benchmark construction pipeline. (a) Discovery and version filtering. (b) EC change detection across consecutive protocol versions. (c) Evidence extraction from protocol PDFs and benchmark assembly.

medically complex participants [31]. Published research provides disease-specific evidence across therapeutic areas including oncology [18, 25], cardiology [36], neurology [21], psychiatry [38], rheumatology [6], and nephrology [4], supported by consensus directives from KDIGO [1] and the Global Cardiovascular Clinical Trialists Forum [7].

Despite this wealth of guidance, no existing system automates the discovery, extraction, and implementation of such recommendations across trials at scale. Individual trialists may consult specific publications when designing or amending protocols, but systematically connecting the large body of published recommendations to active trial designs remains an unsolved technical challenge. The gap is not one of clinical knowledge (the recommendations exist) but of infrastructure: the retrieval and matching between two large bodies of text, published evidence and registered trials, exceeds what manual processes can achieve.

Recent computational work has addressed adjacent problems, including amendment prediction [5] and criteria generation from trial descriptions [23, 32], but none automates evidence-grounded revision (Section 2).

To bridge this gap, we introduce RECITE (**R**evising **E**ligibility **C**riteria **I**ncorporating **T**extual **E**vidence), a framework for evidence-guided eligibility criteria revision. Given original eligibility criteria and supporting documentary evidence, whether from protocol amendment documents, published research papers, or clinical guidelines, RECITE extracts explicit modification directives and generates revised criteria with cited provenance. The framework is designed to operate across diverse evidence sources, handling the variation in structure, length, and style typical of real-world clinical documents.

Our contributions are as follows:

(1) We formulate the task of *evidence-guided eligibility criteria revision*, defining it over diverse documentary sources including protocol amendment documents, published literature, and clinical guidelines.
(2) We develop an agentic literature discovery system that crawls PubMed, matches papers to active trials, and extracts actionable modification directives.
(3) We construct the RECITE benchmark from 567,815 trials on ClinicalTrials.gov, filtering through version histories and protocol evidence to yield 11,913 instances across 5,735 unique trials. We evaluate on a sample of 3,116 instances with original criteria, grounding evidence from protocol amendment documents, and target revisions.
(4) We demonstrate real-world applicability: 44 paper–trial matches yield estimated accrual gains of 25% to 246%, showing that existing clinical knowledge can be operationalized into evidence-based trial design.

2 Background and Related Work

2.1 Eligibility Criteria Design and the Case for Liberalization

Eligibility criteria balance participant safety with scientific validity in clinical trials. Evidence across therapeutic areas suggests that criteria more restrictive than necessary can reduce enrollment, limit demographic diversity, and narrow the generalizability of findings. Oncology has led the evidence base for reform. The ASCO-Friends consensus statement provides concrete recommendations for broadening criteria related to brain metastases, prior malignancies, and organ function thresholds [20], while Trial Pathfinder demonstrated that data-driven relaxation of common exclusions does not compromise hazard ratios [25]. Real-world data analyses have validated these recommendations, showing that applying ASCO-Friends expanded criteria can increase the eligible pool by 78% or more [18]. The FDA's 2020 guidance provides cross-cutting regulatory encouragement for sponsors to broaden criteria and justify each exclusion [31].

This pattern extends beyond oncology: restrictive criteria exclude 63% of lupus patients [6], 86% of depressed outpatients from antidepressant trials [38], and up to 88% of candidates in preclinical Alzheimer's trials [21]. Franzen et al. [9] show that common Alzheimer's exclusions reduce racial and ethnic diversity in enrolled populations, while eligibility complexity in dementia trials has increased over time despite calls for simplification [28]. The proportion of cardiovascular trials excluding chronic kidney disease patients has risen from 66% to 79% over twenty years [4]. Governing bodies have responded with directives from KDIGO [1], the Global Cardiovascular Clinical Trialists Forum [7], and ICH E17 [15]. MDS-specific guidance has called for reducing eligibility barriers in hematologic malignancy trials [3]. Collectively, these

represent explicit, actionable recommendations distributed across governing bodies, academic literature, and institutional protocols.

2.2 Predicting and Implementing Criteria Changes

AMEND++ [5] introduces eligibility criteria amendment prediction as a supervised task, using Change-Aware Masked Language Modeling on historical version diffs spanning 161K trials. Their benchmark provides a resource for studying EC instability over time. RECITE shares the ClinicalTrials.gov data source but addresses a different goal: rather than predicting *whether* amendments will occur, RECITE discovers evidence specifying *what* changes should be made and generates the revised criteria accordingly. The approaches are complementary: AMEND++ identifies trials at high amendment risk; RECITE supplies evidence-grounded revisions. TrialPanorama [33] provides a large-scale trial database with benchmark tasks but does not address evidence-guided revision. Liu et al. [26] demonstrate data-driven eligibility criteria optimization for COVID-19 trials using real-world data, an approach complementary to our evidence-guided revision from published literature.

2.3 AI for Clinical Trial Design and Literature Mining

Generative approaches produce eligibility criteria from trial-level metadata. EC-RAFT [23] uses retrieval-augmented fine-tuning to generate criteria from trial descriptions, while AutoTrial [32] prompts language models for protocol generation. Embedding-based methods have also addressed trial-level representation; for example, Trial2Vec [34] learns self-supervised trial document embeddings for similarity search, and Bettiol et al. [2] cluster eligibility criteria using sentence-level LLM embeddings, achieving 95% of generative-LLM ROUGE scores with compressed cluster representations. These approaches capture distributional patterns across existing criteria. Downstream tasks including patient–trial matching [16] and criteria-to-query translation [29] would benefit from improved criteria. RECITE builds on retrieval-augmented generation [24] and dense passage retrieval [17] to connect published recommendations to trial-level eligibility criteria.

3 Problem Formulation

The task of *evidence-guided eligibility criteria revision* is defined as follows. Given a set of original eligibility criteria E_{orig} for a clinical trial and a documentary evidence source D (which may be a protocol amendment document, a published research paper, a clinical guideline, or any other text containing recommendations for criteria modification), the goal is to produce revised eligibility criteria E_{rev} with provenance. Specifically, a provenance mapping P links each revision to specific passages in D. Formally, the task requires a function $f : (E_{\text{orig}}, D) \rightarrow (E_{\text{rev}}, P)$ that performs two subtasks: (1) extracting explicit modification directives from the documentary text, and (2) faithfully implementing those directives into revised criteria while preserving unmodified elements.

Protocol amendment documents provide naturally occurring ground-truth instances: each yields a triplet of original criteria, evidence document, and revised criteria. These documents vary in length, structure, and detail across institutions, phases, and therapeutic areas, exhibiting the same variation encountered across published papers, guidelines, and institutional reports. Validating on protocol documents therefore tests the core capability required for any evidence source. This framing is supported by prevalence: 75.8% of protocols undergo at least one amendment, and 53% modify eligibility criteria [11, 12].

Revised criteria are evaluated using an LLM-as-judge protocol. GPT-4o has demonstrated strong agreement with human evaluators on text equivalence and quality assessment tasks [37]. The judge (GPT-4o, accessed through UCSF Versa) scores each generated revision on a 0–4 ordinal scale: 0 indicates no overlap with the target revision, 1 indicates minimal overlap, 2 indicates partial overlap with notable gaps, 3 indicates the revision is largely equivalent to the target, and 4 indicates the revision is fully equivalent. The reported metrics are binary equivalence (the proportion of instances scoring ≥ 3, i.e., judged largely or fully equivalent), the normalized score (judge score divided by 4), and, for real-world applications, estimated accrual gains from source literature. Full prompt templates and scoring rubrics are provided in Appendix B.

4 RECITE System

4.1 Framework Overview

RECITE operates through a two-stage pipeline. In Stage 1 (Discovery), the system identifies relevant documentary evidence for a given trial's eligibility criteria by searching published literature, matching papers to trials, and extracting modification directives. In Stage 2 (Revision), given original eligibility criteria and discovered (or provided) evidence, the system generates revised criteria with provenance links to specific evidence passages. For the benchmark evaluation in Section 5, Stage 1 is bypassed because the evidence documents are provided directly. For the real-world applications in Section 6, Stage 1 employs the agentic literature discovery system described below.

The revision stage supports two modes. In full-document mode, the entire evidence document is provided as context alongside the original eligibility criteria. In RAG mode, the evidence document is first chunked and embedded using BAAI/bge-small-en-v1.5 [35] via LlamaIndex, and relevant passages are retrieved via dense passage retrieval with a top-k of 10. Token budgets are enforced using tiktoken's cl100k_base encoding: the query is truncated to 8,192 tokens before embedding, and retrieved passages are fitted within the LLM context window (16,384 tokens for locally hosted models) via LlamaIndex's compact response mode. Both modes produce revised eligibility criteria that cite specific

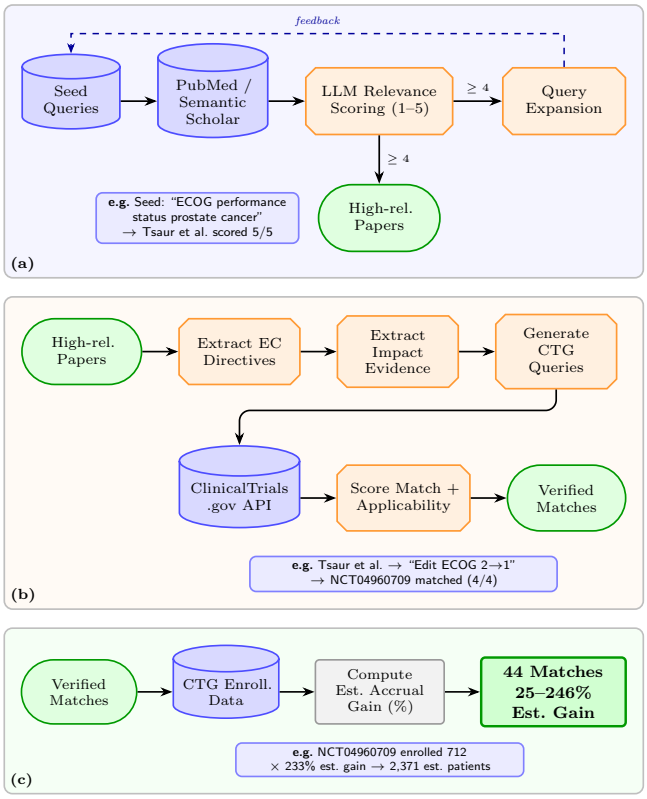

Figure 3: Agentic literature discovery system. (a) Iterative crawl: seed queries retrieve papers from PubMed and Semantic Scholar; an LLM scores relevance (1–5) and high-scoring papers expand the query set. (b) Screening and matching: directives and impact evidence are extracted from each paper, CTG API queries are generated, and candidate (paper, trial) pairs are scored. (c) Accrual estimation: verified matches are joined with CTG enrollment data to compute estimated gains (25–246%).

passages as provenance for each change. This dual-mode design enables full-context processing of shorter evidence documents (such as targeted amendment rationale sections), while RAG mode supports aggregation across longer, multi-source evidence collections such as research papers and clinical guidelines.

4.2 Agentic Literature Discovery

The literature discovery system uses orchestrator-driven iterative query expansion. A Python control loop alternates between LLM-generated search query formulation, adapter-based API search, and LLM-based relevance scoring. The LLM handles only generative steps; all orchestration logic, rate limiting, and state management reside in the Python controller.

The system interfaces with two external APIs through dedicated adapters. The PubMed adapter issues boolean queries via NCBI E-utilities (esearch for identifier retrieval, efetch for metadata) with exponential backoff on transient failures. The Semantic Scholar adapter queries the Graph API, performing citation graph traversal to retrieve both the references and citing papers of each seed (up to 20 in each direction).

In each iteration, the system draws from a seed queue of high-relevance papers managed as a FIFO. If a seed is available, the LLM generates targeted queries informed by the seed's content; otherwise, queries are generated from scratch, avoiding duplicates tracked in a seen-identifier set. Queries are dispatched to PubMed and Semantic Scholar adapters. Each candidate paper is scored on relevance (1–5), extraction confidence (1–5), and accrual impact ease (1–5); papers above the relevance threshold enter the seed queue for progressive scope refinement. The orchestrator terminates after 5 consecutive rounds in which zero new relevant papers are discovered and the seed queue is empty, preventing indefinite cycling.

For paper-trial matching, each paper passing directive extraction (relevance ≥ 4) yields extracted change directives and accrual impact estimates. The LLM generates ClinicalTrials.gov search queries targeting the public API (v2), and each (paper, trial) pair is scored for match quality and applicability (both 1–5). High-quality matches (both ≥ 4) seed additional queries, expanding coverage iteratively. Full scoring prompts and rubrics are provided in Appendix B.

4.3 Evidence-Grounded Revision

The revision pipeline accepts original criteria and evidence, producing revised criteria with passage-level provenance. In RAG mode, evidence documents are chunked, embedded via dense passage retrieval [17], and the top-k passages are provided as context. In full-document mode, the entire document serves as context. Both modes instruct the model to extract explicit directives, apply them, and cite supporting passages.

Directive extraction is validated through fuzzy matching against source text to prevent hallucination. The matching procedure uses a sliding-window approach over the evidence document, comparing each window to the extracted directive using SequenceMatcher-based similarity. Directives with low similarity scores are flagged as potentially unsupported; final validation was performed through manual review against source text.

5 RECITE Benchmark

Protocol amendment documents anchor the RECITE benchmark because each amendment yields a naturally occurring triplet—original eligibility criteria, documentary evidence, and revised criteria—requiring no manual annotation. These documents vary considerably in length, structure, and detail across sponsoring institutions, clinical phases, and therapeutic areas, mirroring the diversity encountered in published papers, clinical guidelines, and institutional reports. Validating directive extraction on this heterogeneous evidence base

therefore tests the core capability required for any documentary source type.

5.1 Data Collection and Curation

The dataset construction begins with 567,815 registered trials from the ClinicalTrials.gov bulk XML download. To identify trials with eligibility criteria modifications, a two-stage heuristic is applied. First, the ClinicalTrials.gov history API reports which modules changed in each version via `moduleLabels` metadata; only versions flagged as modifying the "Eligibility" module are retained. This reduces the candidate set from 567K to 59,996 trials without requiring full text comparison across all version pairs. Second, within this subset, both versions of the eligibility criteria text are normalized (whitespace collapsed, empty lines removed, HTML tags stripped) and compared using a `difflib`-based similarity check. Pairs with similarity above 0.80 or with fewer than 5% of lines changed are discarded as metadata-only or formatting-only differences. Comparing consecutive version pairs across the retained trials yielded 85,083 confirmed eligibility criteria changes across 49,369 unique trials.

To include grounding evidence, trials with downloadable protocol amendment PDFs were identified. The ClinicalTrials.gov API was queried for each of the 49,369 trials with EC changes, checking the largeDocumentModule metadata for protocol availability. Roughly 3,500 trials (approximately 7%) had such documents. Evidence text was extracted from these PDFs using PyMuPDF, then joined with the corresponding eligibility criteria changes to produce complete triplets: original criteria, evidence document, and revised criteria. Amendment-to-version matching uses a date-based heuristic with a 7-day tolerance window; where dates are unavailable, a fallback assumes amendment N corresponds to version N. The resulting dataset contains 11,913 instances across 5,735 trials. A stratified sample of 3,116 instances was selected for benchmarking to manage computational costs: evaluating all eight models under both full-document and RAG conditions requires substantial resources given the mean evidence document length of approximately 134,000 characters. Each instance provides three components: (1) the original eligibility criteria prior to amendment, (2) grounding evidence extracted from the protocol amendment document, and (3) the target revised eligibility criteria as implemented in the subsequent protocol version.

5.2 Dataset Characteristics

Table 1 summarizes the benchmark dataset. The 5,735 unique trials span a broad range of conditions, with breast cancer (98 trials), prostate cancer (57), COVID-19 (56), obesity (53), and heart failure (50) as the most represented. Phase 2 trials account for the largest share (1,663 trials), followed by trials with no reported phase (1,374), Phase 3 (1,059), and Phase 1/2 (439). Trial start dates range from 2008 to 2023. The average evidence document length is approximately 134,000 characters, with wide variation (up to 3.1M characters), reflecting the diversity of protocol documents across institutions and trial phases.

Table 1: RECITE benchmark dataset summary.

Statistic	Value
Total instances	11,913
Unique trials	5,735
Sampled for benchmark	3,116
Avg pre-amendment EC len. (chars)	3,331
Avg amended EC len. (chars)	3,473
Avg evidence len. (chars)	134,180
Trial phases	
Phase 1 / 1–2 / 2	433 / 439 / 1,663
Phase 3 / 4	1,059 / 316
Other / N/A	451 / 1,374
Interventional / observational	5,462 / 272
Trial start date range	2008–2023
Top conditions (unique trials)	
Breast cancer	98
Prostate cancer	57
COVID-19	56
Obesity / Heart failure	53 / 50

Protocol amendment documents differ considerably in length, structure, and level of detail across sponsoring institutions, clinical phases, and therapeutic areas. Some amendments provide tabular summaries of changes with explicit rationale; others embed modification directives within narrative justification spanning multiple pages. This variation mirrors the diversity of evidence sources the framework must handle, since published papers differ in structure from clinical guidelines, which differ from institutional reports.

5.3 Model Evaluation

Eight models were evaluated on the full 3,116-instance sample under both full-document and RAG conditions. The model suite includes six locally hosted open-source models spanning a range of parameter counts: Qwen 2.5 at 0.5B, 3B, and 7B parameters; Gemma 2 at 2B and 9B parameters; and a long-context 7B model (DeepSeek-R1-Distill-Qwen-7B) with a 128K token context window. Two commercial models were also evaluated: GPT-4o (`gpt-4o-2024-08-06`) and GPT-4o-mini, both accessed through UCSF Versa, a PHI-safe, HIPAA-compliant Azure OpenAI deployment. Each model received the full evidence document as context and, separately, a RAG condition where relevant passages were retrieved from the evidence document prior to generation. Table 2 presents results. GPT-4o-mini with RAG achieves the best performance at 85.8% binary equivalence (normalized 0.772). Among open-source models, Gemma 2 9B performs competitively at 84.7% (normalized 0.774) without RAG, within two percentage points of the commercial models.

Scaling effects are pronounced within model families. In the Qwen 2.5 series, binary equivalence improves from 11.1%

at 0.5B parameters to 66.1% at 3B and 81.2% at 7B, indicating that model capacity is a primary determinant of directive extraction and implementation quality. A similar pattern holds for Gemma 2, where performance increases from 37.2% at 2B to 84.7% at 9B. The long-context 7B model achieves only 1.7% binary equivalence despite its 128K context window, suggesting that context length alone does not substitute for model capability on this task.

Table 2: RECITE benchmark results ($n = 3{,}116$). Binary equiv.: fraction of instances where generated and target criteria match (judge score ≥ 3 on a 0–4 ordinal scale). Norm.: mean ordinal judge score normalized to $[0, 1]$. Best per metric in bold.

		no RAG / RAG	
Model	Ctx	Bin. Eq.	Norm.
LongCtx-7B	128K	.017 / .018	.030 / .030
Qwen2.5-0.5B	32K	.111 / .111	.176 / .175
Qwen2.5-3B	32K	.661 / .661	.638 / .638
Qwen2.5-7B	32K	.812 / .809	.741 / .738
Gemma2-2B	8K	.372 / .368	.456 / .456
Gemma2-9B	8K	.847 / .845	**.774** / **.774**
GPT-4o-mini (Versa)	128K	.853 / **.858**	.755 / .772
GPT-4o (Versa)	128K	.853 / .854	.761 / .764

RAG provides modest and inconsistent gains across models. GPT-4o-mini shows a small improvement with RAG (85.3% → 85.8% binary equivalence), while Gemma 2 9B and the Qwen models show negligible differences between conditions. This suggests that for capable models, the full-document setting already provides sufficient signal for directive extraction. The overall finding is that current large language models can extract modification directives from varied documentary evidence and implement them as revised eligibility criteria with high equivalence to target revisions, particularly at the 7B+ parameter scale.

Figure 4 shows that evidence document length has negligible effect on revision quality (Spearman $\rho = -0.04$). The RAG and no-RAG curves are highly correlated (Pearson $R > 0.96$), differing primarily in level rather than trend. In the single-document setting, even documents exceeding 10^6 characters remain within capable models' processing range; retrieval-based methods will become necessary when extending to multi-document evidence sources.

6 Real-World Applications

To demonstrate RECITE beyond the protocol-based benchmark, we applied the full two-stage pipeline to published research literature. The agentic crawling system (Section 4.2) retrieved papers from PubMed that recommend specific eligibility criteria changes, matched them to registered trials on ClinicalTrials.gov, and extracted modification directives with associated accrual impact estimates.

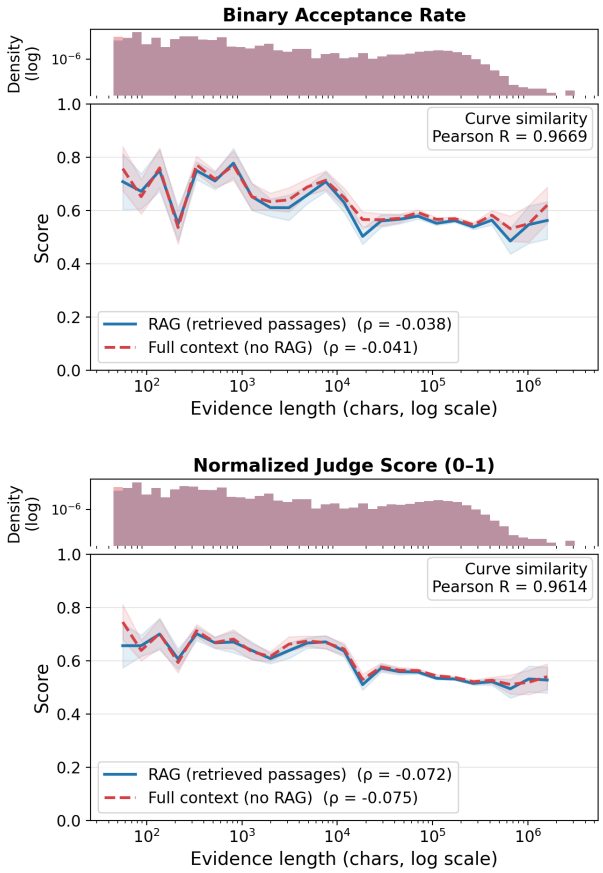

Figure 4: LLM judge performance as a function of evidence document length, pooled across all eight models. (a) Binary equivalence rate. (b) Normalized judge score (0–1). Each panel shows RAG and full-context (no-RAG) curves over 25 log-spaced length bins with 95% CI bands; a marginal density histogram (log-scaled) along the top axis shows the evidence-length distribution.

6.1 Literature–Trial Matching

The system identified 44 paper-trial matches across 6 source publications and 41 unique trials, spanning ASCO-Friends criteria broadening [14, 19], organ function threshold liberalization [27], expanded hepatocellular carcinoma criteria [10], and ECOG performance status adjustments [13, 22]. Each match represents a pairing where the paper's recommended eligibility change is applicable to the trial's current criteria, as determined by the LLM scoring pipeline described in Section 4.2.

Table 3 presents all matches grouped by source paper with current and estimated enrollment. Directives range from targeted threshold changes to broader performance status adjustments, all validated through fuzzy matching against source publications. Match quality scores and rubrics are in Appendix A and Appendix B.

Table 3: Paper–trial matches with extracted directives and estimated enrollment impact. Each group shows a source paper alongside verbatim EC change directives and accrual impact quotes extracted by the pipeline from source publications. Per-trial rows show current and estimated enrollment counts (current × (1 + gain)). M/A: LLM match / applicability scores (1–5 scale).

Source	EC Change Directive	Accrual Impact Quote	Trial (NCT ID)	Enrollment Current →	Enrollment Est.	M / A
Harvey et al. [14], **+100%**	"Change eligibility criteria to allow patients with brain metastases, other malignancies, and a CrCl of ≥ 30 mL/min."	"Broadening three common eligibility criteria has the potential to double the eligible patient population."	Gemcitabine chemo (NCT03101995)	18 →	36	4 / 5
			Durvalumab chemo-immuno (NCT04163432)	43 →	86	4 / 5
Kim et al. [19], **+100%**	"Expand eligibility criteria to allow patients with brain metastases and other malignancies, and lower the CrCl requirement to >30 mL/min."	"Nearly twice as many aNSCLC pts to be considered for trial participation (4851 patients, 46.2%)."	aNSCLC broadening (NCT02792582)	160 →	320	3 / 4
			aNSCLC brain mets (NCT03367819)	44 →	88	3 / 4
			Anti-PD-1 + modulator (NCT04114136)	72 →	144	4 / 5
			aNSCLC eligibility (NCT04488874)	50 →	100	3 / 4
			T-DXd w/ HER2 (NCT04739761)	506 →	1,012	5 / 5
			NK cell therapy (NCT05588453)	30 →	60	4 / 5
			T cells + pembro (NCT06157541)	58 →	116	4 / 5
			Longitudinal biocoll. (NCT06481813)	730 →	1,460	4 / 5
KRAS G12C [27], **+25%**	"Liberalizing criteria for renal dysfunction and brain metastases."	"Would increase enrollment potential by 25% without decreasing the median overall survival of the broadened eligible cohort."	Renal dysfunction + brain mets (NCT01509937)	433 →	541	3 / 3
			CNI replacement (NCT04013620)	48 →	60	4 / 4
			Cancer + BP mgmt (NCT04467021)	61 →	76	4 / 4
			Cerebrovascular device (NCT04704258)	100 →	125	3 / 4
			Moderate renal impair. (NCT05534620)	36 →	45	4 / 4
			EASi-KIDNEY (NCT06531824)	11,000 →	13,750	4 / 4
Fung et al. [10], **+143%**	"Expand eligibility to include patients with slightly worse performance status (ECOG 2) or limited liver dysfunction (Child-Pugh B7)."	"More than twice as many patients would likely receive treatment if MEC were applied."	Abiraterone mCRPC (NCT00485303)	58 →	141	4 / 5
			Belinostat solid tumors (NCT01273155)	72 →	175	5 / 5
			Phase II axitinib (NCT01967576)	14 →	34	4 / 5
			HCC eligibility (NCT02314143)	48 →	116	3 / 4
			Axitinib VTE (NCT03494816)	24 →	58	4 / 4
			First in human YB1-X7 (NCT06889675)	36 →	87	4 / 5
			HCC cabo/rego (NCT06926673)	62 →	150	4 / 4
			HCC eligibility exp. (NCT06984497)	32 →	78	4 / 4
			HCC treatment (NCT06990659)	206 →	500	4 / 4
			HCC cabo/rego exp. (NCT07037394)	252 →	611	4 / 4
ECOG GU [13], **+233%**	"Edit the ECOG score from 2 to 1 for prostate cancer patients and from 3 to 2 for bladder cancer patients."	"Profound increase (233% for prostate cancer patients and 300% for urinary bladder cancer patients) in matched trials per patient."	GU cancer ECOG PS (NCT00428714)	73 →	243	4 / 4
			Abiraterone mCRPC (NCT00485303)	58 →	193	4 / 5
			Phase III radium-223 (NCT00699751)	921 →	3,067	4 / 5
			Phase 2 safety (NCT01756638)	48 →	160	4 / 5
			Bladder cancer ECOG (NCT03768570)	82 →	273	4 / 4
			Darolutamide + ADT Ph. III (NCT04960709)	712 →	2,371	4 / 4
			Darolutamide + ADT (NCT05171387)	78 →	260	4 / 5
			Darolutamide safety (NCT05348876)	50 →	167	4 / 5
			Bladder cancer trial (NCT06524544)	320 →	1,066	4 / 4
			GU ECOG threshold (NCT06661122)	100 →	333	4 / 4
ECOG all [22], **+246%**	"ECOG score per pt should be edited from 3 to 2 or from 2 to 1 to increase the number of matched clinical trials."	"Matched CTs significantly increased by 246% and 189%, respectively, for all cancer types (p<0.001)."	All-cancer ECOG (NCT00022672)	208 →	720	3 / 3
			Abiraterone mCRPC (NCT00485303)	58 →	201	4 / 5
			Phase II axitinib (NCT01967576)	14 →	48	4 / 5
			ECOG-adjusted cancer (NCT03022032)	102 →	353	3 / 4
			Stivant efficacy (NCT03288987)	126 →	436	4 / 5
			Long-term PN (NCT03355079)	2 →	7	4 / 5
			Radiation + immuno (NCT05721755)	290 →	1,003	4 / 4
			ECOG PS threshold (NCT06677255)	149 →	516	3 / 4
Total: 6 papers, 44 matches, estimated gains 25–246%						

6.2 Enrollment Impact Estimates

Estimated accrual gains across the 44 matches range from 25% to 246%, inherited from the source papers. The accrual estimation methodology varies across source publications: some report prospective cohort re-evaluation under broadened criteria [14, 19], while others derive estimates from retrospective analysis of real-world patient data [10, 27]. Gains of 25% correspond to targeted relaxation of renal dysfunction and brain metastasis criteria [27]; the highest gains (246%) correspond to ECOG performance status adjustments ($p < 0.001$) [22]. The system extracts reported percentage gains from the paper text and propagates them to matched trials as scalar estimates (*scalar_gain* = current enrollment × reported percentage gain / 100). These are estimated, not independently verified enrollment outcomes; measuring actual accrual impact would require longitudinal data beyond the current scope.

6.3 Scope and Interpretation

The current application focuses on oncology, reflecting the maturity of EC liberalization research in that area; extension to other therapeutic areas is architecturally straightforward but not yet empirically validated. All matches were among the highest-scoring pairs (see Appendix B); additional before-and-after criteria examples are in Appendix C.

6.4 Illustrative Example

To illustrate how a paper's directive applies to a specific trial, consider Harvey et al. [14] matched to NCT04163432 (Durvalumab chemo-immunotherapy for metastatic non-squamous NSCLC; match 4/5, applicability 5/5). The extracted directive recommends allowing patients with brain metastases, other malignancies, and a creatinine clearance of ≥ 30 mL/min. The original trial already included a renal criterion of eGFR ≥ 30 mL/min, consistent with the paper's CrCl recommendation. However, the trial excluded patients with "active central nervous system metastases and/or carcinomatous meningitis" and "other active malignancy within 2 years except for basal cell carcinoma of the skin or carcinoma in situ of the cervix." Applying the directive broadens both exclusions to permit controlled brain metastases and prior malignancies beyond the narrow exceptions, yielding an estimated 100% accrual gain (43 additional patients).

7 Limitations and Ethical Considerations

Data and benchmark. The RECITE benchmark draws from a specific subset of the trial landscape: of 567,815 registered trials, only those with publicly downloadable protocol amendment PDFs contribute evidence documents, yielding roughly 3,500 trials (0.6% of the registry). Table 1 shows that Phase 2 and Phase 3 trials account for a disproportionate share, and top conditions are weighted toward oncology and well-funded disease areas. The benchmark construction pipeline relies on two key heuristics that trade completeness for efficiency. First, eligibility criteria changes are identified via ClinicalTrials.gov `moduleLabels` metadata rather than exhaustive version-pair comparison across all 567K trials, potentially missing changes where registry metadata is incomplete. Second, text-level change detection uses similarity thresholds that filter out very small substantive changes alongside formatting-only differences. The benchmark validates the framework on protocol documents, and the applications in Section 6 demonstrate transfer to published papers; systematic evaluation across the full range of document types (e.g., clinical guidelines, institutional reports) remains future work. The LLM-as-judge evaluation was validated through manual inspection of scored instances. The equivalence-judging task—determining whether two eligibility criteria texts are functionally identical—is well-suited to LLM evaluation given its reliance on structured text comparison. However, LLM judges may exhibit systematic biases related to hallucination or scoring precision, and inter-rater reliability against human expert judgment has not been established at scale.

Applications. The 44 paper–trial matches represent the highest-scoring pairs within the scoring system (see Appendix B), not a random sample. Estimated accrual gains (25–246%) are inherited directly from the source publications' reported findings. This delegation to peer-reviewed evidence is by design: RECITE surfaces and operationalizes existing research conclusions rather than generating independent enrollment predictions, grounding impact estimates in validated study results. These estimates were not independently verified against actual enrollment data. The current applications focus on oncology, and directive extraction operates on abstracts rather than full text, meaning more detailed recommendations in methods or results sections may be missed.

Ethics. RECITE is a decision-support tool, not a replacement for clinical judgment. All proposed revisions require review by clinical investigators, IRB approval, and regulatory review where applicable [31]. The system provides provenance linking each revision to specific source passages to support expert evaluation. The published literature from which RECITE draws recommendations may itself reflect biases in what is studied; while broadening eligibility criteria generally improves demographic diversity [18], RECITE does not explicitly optimize for representativeness.

Future work. Key directions include extending directive extraction to full paper text (beyond abstracts), which would capture more detailed recommendations in methods and results sections. Constructing a literature-linked benchmark from protocol document references would complement our protocol-based benchmark with a paper-based evaluation setting. Matching modified criteria to actual enrollment outcomes using longitudinal registry data would provide ground-truth validation of the accrual impact estimation pipeline. Temporal meta-analysis of how eligibility criteria evolve in response to published evidence would further establish the relationship between evidence publication and protocol modification.

8 Conclusion

RECITE provides infrastructure to discover, extract, and implement evidence-based eligibility criteria revisions with full provenance. On a benchmark of 3,116 instances derived from protocol amendments on ClinicalTrials.gov, the best configuration achieves 85.8% binary equivalence under an LLM-as-judge protocol. Applied to published literature through an agentic discovery system, RECITE identifies 44 paper–trial matches with estimated accrual gains of 25–246%. Both evidence types, protocol amendments and published papers, share the property that makes this task tractable: explicit modification directives embedded in varied text. The underlying pattern of extracting directives and implementing them in design documents could generalize to other trial elements such as endpoint selection or dosing regimens.

Acknowledgments

This work was supported in part by the National Science Foundation Graduate Research Fellowship Program. We thank the UCSF Versa team for providing PHI-safe access to Azure OpenAI models. **Code, benchmark construction resources, and supplementary materials are available at https://github.com/abbasilab/RECITE.**

GenAI Disclosure

Research methodology. Large language models are the primary subject of this work. Eight LLM configurations (GPT-4o, GPT-4o-mini, Gemma 2 2B/9B, Qwen 2.5 0.5B/3B/7B, and DeepSeek-R1-Distill-Qwen-7B) were evaluated as benchmark subjects. GPT-4o served as the LLM-as-judge evaluator (Section 3). LLMs were also used within the agentic literature discovery pipeline for search query formulation, relevance scoring, and directive extraction (Section 4.2). All methodological uses are described in the paper.

Manuscript preparation. Claude (Anthropic) was used to assist with editing, restructuring, and refining manuscript prose. All generated or modified text was reviewed, verified, and approved by the authors, who take full responsibility for the final content.

Code development. Claude (Anthropic) was used for code completion and debugging support during development of the RECITE framework. All code was reviewed and tested by the authors.

References

[1] Colin Baigent, William G Herrington, Josef Coresh, Martin J Landray, Adeera Levin, Vlado Perkovic, et al. 2017. Challenges in conducting clinical trials in nephrology: conclusions from a Kidney Disease—Improving Global Outcomes (KDIGO) Controversies Conference. *Kidney International* 92, 2 (2017), 297–305. doi:10.1016/j.kint.2017.04.019 "Eligibility criteria must be practical and broad".

[2] Alberto Bettiol et al. 2025. Analysis of eligibility criteria clusters based on large language models for clinical trial design. *Journal of the American Medical Informatics Association* 32, 3 (2025), 447–458. doi:10.1093/jamia/ocae311 LLM-embedding clustering of EC for protocol generation.

[3] Uma Borate, Kelly Pugh, Allyson Waller, Rina Li Welkie, Ying Huang, Jan Philipp Bewersdorf, Maximilian Stahl, Amy E DeZern, Uwe Platzbecker, Mikkael A Sekeres, et al. 2025. Reducing clinical trial eligibility barriers for patients with myelodysplastic syndromes. *Blood* 145, 13 (2025), 1369. https://ashpublications.org/blood/article/145/13/1369/534836/Reducing-clinical-trial-eligibility-barriers-for Non-oncology EC liberalization example in hematology/MDS.

[4] Julia M T Colombijn, Demy L Idema, Sanne van Beem, et al. 2024. Representation of Patients With Chronic Kidney Disease in Clinical Trials of Cardiovascular Disease Medications: A Systematic Review. *JAMA Network Open* 7, 3 (2024), e240427. doi:10.1001/jamanetworkopen.2024.0427 CKD exclusion rose from 66% to 79% over two decades.

[5] Trisha Das, Mandis Beigi, Jacob Aptekar, and Jimeng Sun. 2025. AMEND++: Benchmarking Eligibility Criteria Amendments in Clinical Trials. arXiv:2601.06300 Binary classification: predicts WHETHER amendments occur.

[6] Sarah Dyball, Sophie Collinson, Emily Sutton, Eoghan M McCarthy, Ian N Bruce, and Ben Parker. 2021. Lupus clinical trial eligibility in a real-world setting: results from the British Isles Lupus Assessment Group-Biologics Register (BILAG-BR). *Lupus Science & Medicine* 8 (2021), e000513. doi:10.1136/lupus-2021-000513 63% of SLE patients ineligible for non-renal Phase III trials.

[7] Laura Filbey, Harriette G C Van Spall, et al. 2023. Improving representativeness in trials: a call to action from the Global Cardiovascular Clinical Trialists Forum. *European Heart Journal* 44, 11 (2023), 921–930. doi:10.1093/eurheartj/ehac788

[8] David B Fogel. 2018. Factors associated with clinical trials that fail and opportunities for improving the likelihood of success: A review. *Contemporary Clinical Trials Communications* 11 (2018), 156–164. doi:10.1016/j.conctc.2018.08.001 80% of trials fail enrollment timelines; 48% of sites under-enroll.

[9] Sanne Franzen, Jade Emily Smith, Esther van den Berg, Monica Rivera Mindt, et al. 2022. Diversity in Alzheimer's disease drug trials: The importance of eligibility criteria. *Alzheimer's & Dementia* 18, 4 (2022), 810–823. doi:10.1002/alz.12433

[10] Alan S Fung, Vincent C Tam, Daniel E Meyers, Hao-Wen Sim, Jennifer J Knox, et al. 2019. Real world eligibility for cabozantinib (C), regorafenib (Reg), and ramucirumab (Ram) in hepatocellular carcinoma (HCC) patients after sorafenib (S). In *Journal of Clinical Oncology*, Vol. 37. 422. doi:10.1200/JCO.2019.37.4_SUPPL.422

[11] Kenneth Getz, Zachary Smith, Emily Botto, Elisabeth Murphy, and Arnaud Dauchy. 2024. New Benchmarks on Protocol Amendment Practices, Trends and their Impact on Clinical Trial Performance. *Therapeutic Innovation & Regulatory Science* 58, 3 (2024), 539–548. doi:10.1007/s43441-024-00622-9 75.8% prevalence; mean 3.3 amendments; 260-day implementation.

[12] Kenneth A Getz, Stella Stergiopoulos, Mary Short, Leon Surgeon, Randy Krauss, Sybrand Pretorius, Julian Desmond, and Derek Dunn. 2016. The Impact of Protocol Amendments on Clinical Trial Performance and Cost. *Therapeutic Innovation & Regulatory Science* 50, 4 (2016), 436–441. doi:10.1177/2168479016632271 57% of protocols amended; 53% modify EC; $535K per Phase III amendment.

[13] Liat Gortzak Uzan, Liran Lapidot, Tal Bader, and Anat Gaziel. 2023. ECOG score as a barrier for clinical trial eligibility in the patient population with genitourinary cancer: Is it time to adapt this criterion to real-world data?. In *Journal of Clinical Oncology*, Vol. 41. 78. doi:10.1200/jco.2023.41.6_suppl.78

[14] R. Donald Harvey, Suanna S Bruinooge, Li Chen, Elizabeth Garrett-Mayer, Whitney Rhodes, Edward Stepanski, Thomas S Uldrick, Gwynn Ison, Sean Khozin, William S Rubinstein, Charles Schenkel, Richard S Miller, George A Komatsoulis, Richard L Schilsky, and Edward S Kim. 2021. Impact of Broadening Trial Eligibility Criteria for Patients with Advanced Non–Small Cell Lung Cancer: Real-World Analysis of Select ASCO-Friends Recommendations. *Clinical Cancer Research* 27, 9 (2021), 2430–2434. doi:10.1158/1078-0432.CCR-20-3857

[15] International Council for Harmonisation. 2017. General Principles for Planning and Design of Multi-Regional Clinical Trials (E17). https://www.fda.gov/media/99974/download. Addresses eligibility harmonization across regions.

[16] Qiao Jin, Zifeng Wang, Charalampos S Floudas, Fangyuan Chen, Changlin Gong, Dara Bracken-Clarke, Elisabetta Xue, Yifan Yang, Jimeng Sun, and Zhiyong Lu. 2024. TrialGPT: Matching Patients to Clinical Trials with Large Language Models. *Nature Communications* 15, 1 (2024), 9074. doi:10.1038/s41467-024-53081-z Patient-trial matching downstream of EC.

[17] Vladimir Karpukhin, Barlas Oğuz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. Dense Passage Retrieval for Open-Domain Question Answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*. 6769–6781. doi:10.18653/v1/2020.emnlp-main.550

[18] Maneet Kaur, Filip Frahm, Yichen Lu, Mustafa S Ascha, Jenny S Guadamuz, Efrat Dotan, Adam S Gottesman, et al. 2024. Broadening Eligibility Criteria and Diversity among Patients for Cancer Clinical Trials. *NEJM Evidence* 3, 4 (2024), EVIDoa2300236. doi:10.1056/EVIDoa2300236 78% increase in eligible pool with broadened criteria.

[19] Edward S Kim et al. 2019. Impact of broadening clinical trial eligibility criteria for advanced non-small cell lung cancer patients: Real-world analysis. In *Journal of Clinical Oncology*, Vol. 37. LBA108. doi:10.1200/JCO.2019.37.18_SUPPL.LBA108

[20] Edward S Kim, Suanna S Bruinooge, Sarah Roberts, Gladys Berénos-Riley, Robert Lorence, Thomas Mack, Richard L Schilsky, Sigal A Wagner, James R Rigas, Patrick J Loehrer, et al. 2021. Broadening Eligibility Criteria to Make Clinical

Trials More Representative: American Society of Clinical Oncology and Friends of Cancer Research Joint Research Statement. *Journal of Clinical Oncology* 39, 4 (2021), 370–380. doi:10.1200/JCO.20.02636 ASCO-Friends recommendations for brain mets, prior malignancies, organ function.

[21] Jessica B Langbaum et al. 2023. Recommendations to address key recruitment challenges of Alzheimer's disease clinical trials. *Alzheimer's & Dementia* 19, 2 (2023), 696–707. doi:10.1002/alz.12737 Screen-failure rates reach 88% in preclinical AD trials.

[22] Yael Lapidot, Merav Safran, Tal Bader, and Anat Gaziel. 2023. Profound impact of ECOG performance status on clinical trial eligibility and access: Diverging effects by cancer type. In *JCO Global Oncology*, Vol. 9. 97. doi:10.1200/go.2023.9.supplement_1.97

[23] Nopporn Lekuthai, Nattawit Pewngam, Supitcha Sokrai, and Titipat Achakulvisut. 2025. EC-RAFT: Automated Generation of Clinical Trial Eligibility Criteria through Retrieval-Augmented Fine-Tuning. In *Findings of the Association for Computational Linguistics: ACL 2025*. 9432–9444. doi:10.18653/v1/2025.findings-acl.491 Retrieval-augmented fine-tuning for EC generation from trial descriptions.

[24] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. 2020. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In *Advances in Neural Information Processing Systems*, Vol. 33. 9459–9474.

[25] Ruishan Liu, Shemra Rizzo, Samuel Whipple, Navdeep Pal, Arturo Lopez Pineda, Michael Lu, Brandon Arnieri, Ryan Copping, Marius Garmhausen, and James Zou. 2021. Evaluating eligibility criteria of oncology trials using real-world data and AI. *Nature* 592, 7855 (2021), 629–633. doi:10.1038/s41586-021-03430-5 Trial Pathfinder: data-driven EC optimization.

[26] Sining Liu, Henry T Hepburn, Ali Khalili, et al. 2021. Towards clinical data-driven eligibility criteria optimization for interventional COVID-19 clinical trials. *Journal of the American Medical Informatics Association* 28, 9 (2021), 1993–2001. doi:10.1093/jamia/ocaa295 Relaxing eGFR 60→30 increased eligibility from 56% to 81%.

[27] Meaghan S May, Mark Wooster, Prashasti Agrawal, Jae Won Lee, Brandon May, Xiaomei Ma, et al. 2025. Multicenter study of the impact of trial eligibility criteria on enrollment to KRAS G12C inhibitor trials in patients with non-small cell lung cancer. In *Journal of Clinical Oncology*, Vol. 43. 11171. doi:10.1200/jco.2025.43.16_suppl.11171

[28] Ashlee K Mitchell et al. 2024. Analysis of eligibility criteria in Alzheimer's and related dementias clinical trials. *Scientific Reports* 14 (2024), 15234. doi:10.1038/s41598-024-65767-x

[29] Jimyung Park, Yilu Fang, Casey Ta, Gongbo Zhang, Betina Idnay, Fangyi Chen, David Feng, Rebecca Shyu, Emily R Schriver, and Chunhua Weng. 2024. Criteria2Query 3.0: Leveraging generative large language models for clinical trial eligibility query generation. *Journal of Biomedical Informatics* 154 (2024), 104649. doi:10.1016/j.jbi.2024.104649 Maps EC to executable database queries.

[30] Olesya Razuvayevskaya, Irene Lopez, Ian Dunham, and David Ochoa. 2024. Genetic factors associated with reasons for clinical trial stoppage. *Nature Genetics* 56 (2024), 1862–1867. doi:10.1038/s41588-024-01854-z 28,561 stopped trials; insufficient enrollment is #1 reason at 36.67%.

[31] U.S. Food and Drug Administration. 2020. Enhancing the Diversity of Clinical Trial Populations: Eligibility Criteria, Enrollment Practices, and Trial Designs: Guidance for Industry. https://www.fda.gov/regulatory-information/search-fda-guidance-documents/enhancing-diversity-clinical-trial-populations-eligibility-criteria-enrollment-practices-and-trial. Cross-cutting guidance: broaden EC, justify each exclusion.

[32] Zifeng Wang et al. 2023. AutoTrial: Prompting Language Models for Automated Clinical Trial Design. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. LLM-based protocol generation without external evidence grounding.

[33] Zifeng Wang et al. 2025. TrialPanorama: A Large-Scale, Diverse, and Fine-Grained Clinical Trial Database and Benchmark. arXiv:2502.08950 Large-scale trial database and benchmark tasks.

[34] Zifeng Wang and Jimeng Sun. 2022. Trial2Vec: Zero-Shot Clinical Trial Document Similarity Search using Self-Supervision. In *Findings of the Association for Computational Linguistics: EMNLP 2022*. Association for Computational Linguistics, 6377–6390. doi:10.18653/v1/2022.findings-emnlp.476 Self-supervised trial embeddings for similarity search.

[35] Shitao Xiao, Zheng Liu, Peitian Zhang, Niklas Muennighoff, Defu Lian, and Jian-Yun Nie. 2024. C-Pack: Packaged Resources To Advance General Chinese Embedding. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*. ACM, 641–650. doi:10.1145/3626772.3657878 Introduces BGE embedding model family including bge-small-en-v1.5.

[36] Faiez Zannad, Harriette G C Van Spall, et al. 2024. How to make cardiology clinical trials more inclusive. *Nature Medicine* 30, 10 (2024), 2745–2755. doi:10.1038/s41591-024-03273-3

[37] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2024. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems*, Vol. 36. 46595–46623.

[38] Mark Zimmerman, Caroline Balling, Iwona Chelminski, and Kristy Dalrymple. 2019. Have Treatment Studies of Depression Become Even Less Generalizable? Applying the Inclusion and Exclusion Criteria in Placebo-Controlled Antidepressant Efficacy Trials Published over 20 Years to a Clinical Sample. *Psychotherapy and Psychosomatics* 88, 3 (2019), 165–170. doi:10.1159/000499917 86.1% mean exclusion rate; worsening over time (91.4% for 2010–2014).

A Full Paper–Trial Matches

All 44 paper–trial matches with enrollment impact estimates, source papers, extracted directives, and LLM evaluation scores are presented in Table 3. For each match, the Match score (M, 1–5) assesses topical alignment between the paper's recommendations and the trial's eligibility criteria, and the Applicability score (A, 1–5) measures how directly the paper's directive can be implemented in the trial. Scoring rubrics and prompt templates are provided in Appendix B.

B Scoring Prompts and Rubrics

This appendix documents the LLM scoring prompts, rubrics, thresholds, and filtering logic used throughout the RECITE pipeline. All prompts are stored in JSON configuration files and loaded at runtime. Full prompt templates are available in the code repository.

B.1 Paper Relevance Scoring

Each retrieved paper is scored on three dimensions (1–5 each). Table 4 shows the relevance rubric; extraction confidence and accrual ease follow analogous scales.

Extraction confidence measures how actionable the paper's recommendations are, ranging from 1 (no actionable information) to 5 (directly applicable threshold changes such as "eGFR <30 → <20" or "platelets ≥150k → ≥75k"). **Accrual ease** measures how quantifiable the enrollment impact is, ranging from 1 (vague or subjective criteria) to 5 (easily quantifiable biomarkers like age or blood pressure).

The LLM returns a JSON object with all three scores and brief reasoning. Papers with relevance ≥ 3 proceed to directive extraction. The system also supports configurable filtering on extraction confidence and accrual ease.

Table 4: Paper relevance scoring rubric.

Score	Definition
1	Not relevant: does not discuss eligibility criteria, enrollment, or trial participation
2	Tangentially relevant: mentions clinical trials or enrollment but not specifically about criteria
3	Somewhat relevant: discusses eligibility criteria but not central focus
4	Highly relevant: focuses on eligibility criteria, enrollment barriers, or trial participation issues
5	Directly relevant: directly about eligibility criteria modification, broadening, loosening, or specific criteria changes

B.2 Paper–Trial Match Scoring

Each (paper, trial) pair is scored on two dimensions. Table 5 shows both rubrics.

Table 5: Paper–trial match and applicability scoring rubrics.

Score	Match (topic alignment)	Applicability (ease of application)
1	No meaningful alignment; different conditions, no EC overlap	Requires deep digging into full paper; change is implied but not explicit
2	Minimal alignment; may share broad topic but differ significantly	Hints at change but vague; may require reading full paper
3	Some alignment with gaps; similar condition, some EC overlap	Change is mentioned but not overt; requires some interpretation
4	Good alignment with minor gaps; same or similar condition, substantial EC overlap	Directive is present but may need minor interpretation
5	Strong alignment across all dimensions; same condition, significant EC overlap	Right there in abstract and very overt; can be applied immediately

The scorer also extracts verbatim **change directive quotes** (imperative mood, e.g., "change X to Y") and **change rationale quotes** from the paper. High-quality matches (both scores ≥ 4) are used as seeds for additional query generation. The default filtering threshold is match ≥ 3, with results sorted by match score descending, then applicability score descending. A maximum of 10 matches per paper are retained.

B.3 LLM Judge (Benchmark Evaluation)

The benchmark judge scores each (target, predicted) eligibility criteria pair on two scales: a **binary score** (0 = incorrect, 1 = correct) and an **ordinal score** on a 0–4 scale where 0 indicates no correspondence with the target, 1 indicates poor correspondence with major omissions, 2 indicates partial correspondence with notable gaps, 3 indicates good correspondence with minor differences, and 4 indicates the prediction is essentially identical to the target. We report binary acceptability as the proportion of instances with ordinal score ≥ 3. For efficiency, predictions are evaluated in batches, with the judge returning a JSON dictionary mapping instance indices to score pairs.

B.4 Directive and Impact Extraction

The directive extraction prompt classifies each paper as containing exact directives (extractable from abstract), directives requiring full text, or no directives. Only papers with extractable directives proceed. The impact extraction prompt identifies numeric accrual estimates (percent increase or absolute patient counts) with supporting verbatim evidence. Key computed quantities are: `paper_pct_gain` (percent increase in eligible patients, e.g., 100 = doubling) and `scalar_gain` (estimated additional patients for a given trial = enrollment $\times$ paper_pct_gain / 100).

B.5 Query Generation

Three query generation prompts are used: (1) literature search queries for PubMed and Semantic Scholar, (2) trial matching queries for ClinicalTrials.gov derived from paper content, and (3) seed expansion queries generated from high-quality matches. All prompts receive previously issued queries to avoid repetition and return JSON arrays of query strings.

C Representative Directive Applications

Below are six representative examples showing how RECITE applies a paper's directive to a matched trial's eligibility criteria. Revised criteria were generated by applying each paper's extracted directive to the original trial EC text. The full set of 44 applications is available in the code repository.

C.1 Example 1: ASCO-Friends Broadening Applied to Durvalumab Trial

Source: Harvey et al. [14] (Match: 4/5, Applicability: 5/5)
Trial: NCT04163432 (Durvalumab chemo-immunotherapy, metastatic non-squamous NSCLC)
Directive: "Change eligibility criteria to allow patients with brain metastases, other malignancies, and a creatinine clearance (CrCl) of $\geq$ 30 mL/minute."
Estimated accrual gain: 100% (43 additional patients)
Key change: The original trial already included a renal criterion of eGFR $\geq$ 30 mL/min, consistent with the paper's CrCl recommendation. The directive additionally recommends removing exclusions for controlled brain metastases and prior non-lung malignancies. The original exclusion criteria included "active central nervous system metastases

and/or carcinomatous meningitis" and "other active malignancy within 2 years except for basal cell carcinoma of the skin or carcinoma in situ of the cervix." Applying the directive broadens both to permit controlled brain metastases and prior malignancies beyond the narrow exceptions.

C.2 Example 2: ECOG Adjustment Applied to Abiraterone Trial

Source: Tsaur et al. [13] (Match: 4/5, Applicability: 5/5)
Trial: NCT00485303 (Abiraterone acetate, metastatic castration-resistant prostate cancer)
Directive: "Edit the ECOG score from 2 to 1 for prostate cancer patients."
Estimated accrual gain: 233% (135 additional patients)

Original criterion: "Eastern Cooperative Oncology Group (ECOG) Performance Status of less than or equal to 2 (Karnofsky Performance Status $\geq$ 50%)"

Revised criterion: "Eastern Cooperative Oncology Group (ECOG) Performance Status of less than or equal to 1 (Karnofsky Performance Status $\geq$ 70%)"

This is a single-line change: the ECOG threshold moves from $\leq$2 to $\leq$1. While this appears to *tighten* rather than loosen the criterion, the source paper's finding is that restricting to ECOG $\leq$1 dramatically increases the number of trials a patient matches (233% increase), because the majority of prostate cancer trials already require ECOG $\leq$1. The directive's value is in identifying the mismatch between this trial's ECOG $\leq$2 requirement and the broader trial landscape, where harmonizing criteria could improve cross-trial patient matching and accrual.

C.3 Example 3: ASCO-Friends Broadening Applied to T-DXd HER2 Trial

Source: Kim et al. [19] (Match: 5/5, Applicability: 5/5)
Trial: NCT04739761 (Trastuzumab deruxtecan in participants with or without HER2-positive breast cancer and brain metastases)
Directive: "Expand eligibility criteria to allow patients with brain metastases and other malignancies, and lower the creatinine clearance requirement to >30 mL/min."
Estimated accrual gain: 100% (506 additional patients)

Key change: This trial already permits patients with untreated, stable, or progressing brain metastases and requires ECOG 0–1. The directive's brain metastasis recommendation is thus already partially satisfied. The primary broadening applies to the "other malignancies" exclusion and the creatinine clearance threshold. This represents a high-concordance match (5/5) where the trial's existing design already aligns with part of the ASCO-Friends recommendations, and the directive identifies remaining restrictions that could be relaxed.

C.4 Example 4: Renal and Brain Metastasis Liberalization Applied to KRAS G12C Trial

Source: KRAS G12C multicenter study [27] (Match: 3/5, Applicability: 3/5)
Trial: NCT01509937 (Phase I/II kidney transplant study with renal dysfunction criteria)
Directive: "Liberalizing criteria for renal dysfunction and brain metastases."
Estimated accrual gain: 25% (108 additional patients)

Key change: The moderate match score (3/5) reflects that this trial addresses renal dysfunction in a transplant setting rather than the original NSCLC context. The directive's recommendation to liberalize renal function thresholds is still applicable: relaxing creatinine clearance or eGFR cutoffs could increase the eligible pool. This example illustrates how RECITE identifies cross-specialty applicability of liberalization recommendations, even when the therapeutic context differs from the source paper.

C.5 Example 5: HCC Performance Status Expansion Applied to Belinostat Trial

Source: Fung et al. [10] (Match: 5/5, Applicability: 5/5)
Trial: NCT01273155 (Belinostat for solid tumors and lymphomas, Phase I/II)
Directive: "Expand eligibility to include patients with slightly worse performance status (ECOG 2) or limited liver dysfunction (Child-Pugh B7)."
Estimated accrual gain: 143% (103 additional patients)

Original criterion: "Eastern Cooperative Oncology Group (ECOG) performance status less than or equal to 2 (Karnofsky greater than or equal to 60 percent)."

Key change: This trial already allows ECOG $\leq$2, so the performance status component of the directive is already satisfied. The primary applicable change is the Child-Pugh criterion: the original trial requires adequate hepatic function but does not explicitly address Child-Pugh classification. Expanding to include Child-Pugh B7 patients, who have limited liver dysfunction, would broaden enrollment for this solid tumor study. The 5/5 match score reflects the strong alignment between the directive's liver dysfunction recommendations and the trial's hepatic function requirements.

C.6 Example 6: Pan-Cancer ECOG Adjustment Applied to Stivant Efficacy Trial

Source: Tsaur et al. [22] (Match: 4/5, Applicability: 5/5)
Trial: NCT03288987 (Stivant efficacy for metastatic colorectal cancer)
Directive: "ECOG score per pt should be edited from 3 to 2 or from 2 to 1 to increase the number of matched clinical trials."
Estimated accrual gain: 246% (310 additional patients)

Original criterion: "Eastern Cooperative Oncology Group (ECOG) performance status of ≤ 1."

Key change: This trial already requires ECOG $\leq$1. The directive recommends editing ECOG thresholds (3→2 or 2→1) across cancer types to increase cross-trial matching. For this trial, the relevant interpretation is that the strict ECOG $\leq$1 requirement, while common, contributes to a pattern where patients with ECOG 2 are systematically excluded from a large fraction of trials. The source paper finds that this threshold adjustment increases the number of matched trials per patient by 246% across all cancer types ($p < 0.001$). The high applicability score (5/5) reflects that the trial's ECOG requirement is exactly the type of criterion the directive targets.